

CAD Modeling and Nonlinear Finite Element Analysis of a Modular Pneu-Net Soft Pneumatic Actuator

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Abstract

This project presents the CAD modeling and nonlinear finite element analysis of a modular Pneu-Net soft pneumatic actuator inspired by the design concept reported in “*Modular Soft Pneumatic Actuator Mimics Elephant Trunk Locomotion*.” The study focuses on pressure-induced bending behavior and investigates the influence of actuator wall thickness on deformation, strain response, and numerical behavior. The three-dimensional actuator geometry was modeled in SolidWorks and transferred to Abaqus/Standard for nonlinear static analysis. The elastomeric material was represented using a second-order Yeoh hyperelastic model, while geometric nonlinearity was included to account for large deformations. Quadratic hybrid tetrahedral elements were employed to accommodate the nearly incompressible material behavior. Two wall-thickness configurations, 2.0 mm and 2.5 mm, were investigated under multiple internal-pressure cases. The simulations demonstrated a clear pressure-dependent bending response and showed that wall thickness significantly influences actuator compliance and numerical robustness. The developed computational workflow provides a practical basis for preliminary design evaluation and further development of modular soft pneumatic actuators for assistive and rehabilitation-oriented applications.

Keywords: Soft Pneumatic Actuator; Pneu-Net; Soft Robotics; Hyperelasticity; Yeoh Model; Nonlinear Finite Element Analysis; Abaqus

1. Introduction

Soft robotics has emerged as an important area of robotics and mechatronics due to its ability to produce compliant, adaptive, and biologically inspired motion. Unlike conventional rigid mechanisms, soft robotic systems employ highly deformable materials that can undergo large strains while maintaining safe and flexible interaction with their environment. These characteristics make soft actuators particularly attractive for applications involving direct human interaction, including rehabilitation systems, wearable assistive devices, exoskeletons, and hand-assist mechanisms [2,5,6].

Among the different actuation strategies used in soft robotics, soft pneumatic actuators are widely studied because of their lightweight structure, large deformation capability, and relatively simple actuation principle. In Pneu-Net actuators, internal pneumatic chambers are pressurized to generate controlled deformation, while the geometry of the chamber network, material properties, and applied pressure determine the resulting bending behavior. Accurate numerical modeling of such actuators requires the consideration of both material and geometric nonlinearities. Elastomeric materials are commonly represented using hyperelastic constitutive models, while large-deformation finite element analysis is required to capture the

nonlinear response under internal pressure. Numerical stability is also an important consideration because nearly incompressible materials and highly deformable geometries can introduce convergence difficulties [3,7].

In this project, a modular Pneu-Net soft pneumatic actuator was modeled and analyzed based on the design concept reported in “*Modular Soft Pneumatic Actuator Mimics Elephant Trunk Locomotion*.” The actuator geometry was developed in SolidWorks and subsequently analyzed in Abaqus/Standard using nonlinear finite element analysis. The primary objective was to investigate the effect of wall thickness on pressure-induced bending, deformation, strain response, and numerical behavior. Two wall-thickness configurations, 2.0 mm and 2.5 mm, were considered and compared under multiple internal-pressure cases.

2. Literature Background

Pneu-Net soft pneumatic actuators consist of interconnected internal chambers that deform under pneumatic pressurization. Their mechanical response is strongly influenced by chamber geometry, material properties, wall thickness, and the applied pressure level. Previous studies have shown that geometric parameters and chamber

arrangement can be tailored to control the magnitude and direction of actuator deformation, while reinforcement strategies may be used to limit undesired strain modes [2–4].

The principal design inspiration for this project was the modular soft pneumatic actuator proposed by Elchrif et al. [1]. Their bio-inspired architecture was developed to reproduce motion patterns similar to elephant-trunk locomotion and to enable multiple finger-motion directions. The reported actuator employs a modular arrangement of pneumatic networks whose independent or combined pressurization can generate different bending patterns and multiple degrees of freedom. The reference study also employed nonlinear finite element analysis to model the large-deformation response of the actuator. A second-order Yeoh hyperelastic model was used to describe the elastomeric material, while the actuator base was constrained and internal pressure was applied to the pneumatic chamber surfaces. These concepts provided the main design and modeling basis for the present project, in which a related modular Pneu-Net geometry was developed in SolidWorks and evaluated numerically in Abaqus/Standard, with particular emphasis on wall-thickness effects and pressure-induced bending.

3. Actuation Strategy

The bending behavior of the actuator was defined through the selection of active pneumatic networks and the applied internal-pressure profile. Because deformation in a Pneu-Net actuator is generated by pressurizing its internal chambers, different actuation configurations can be used to produce different bending directions and magnitudes.

In the modular architecture considered in this project, pneumatic networks may be activated individually or in combination. Pressurizing a single network produces directional bending, whereas simultaneous pressurization of adjacent networks can generate a stronger and more stable bending response. The investigated simulation cases primarily focused on pressurization of adjacent pneumatic networks to evaluate the actuator response under representative bending conditions.

Internal pressure was applied progressively during the nonlinear analysis to avoid abrupt loading and to improve numerical stability. The resulting actuator response was evaluated using deformation shape, tip displacement, and strain-related outputs. These quantities were subsequently used to compare the behavior of the different wall-thickness configurations.

4. Actuator Design and CAD Modeling

The actuator geometry was developed in SolidWorks using a modular Pneu-Net architecture inspired by the reference design described in Section 2. A base module containing the internal pneumatic chamber network was modeled first, after which repeated modules were assembled to form the complete actuator geometry. The internal cavities, pneumatic channels, and pressure-inlet region were incorporated into the three-dimensional model to enable subsequent pressure-based finite element analysis.

Two geometric configurations were prepared with nominal wall thicknesses of 2.0 mm and 2.5 mm. The remaining overall architecture was kept comparable between the two variants so that the influence of wall thickness on the actuator response could be investigated. The CAD geometry was also checked for geometric continuity and interference to facilitate transfer to the finite element environment.

After completion of the CAD models, the actuator geometries were exported in STEP format for import into Abaqus/CAE. The imported geometry was subsequently prepared as a single analysis part for material assignment, meshing, boundary-condition definition, and internal-pressure loading.

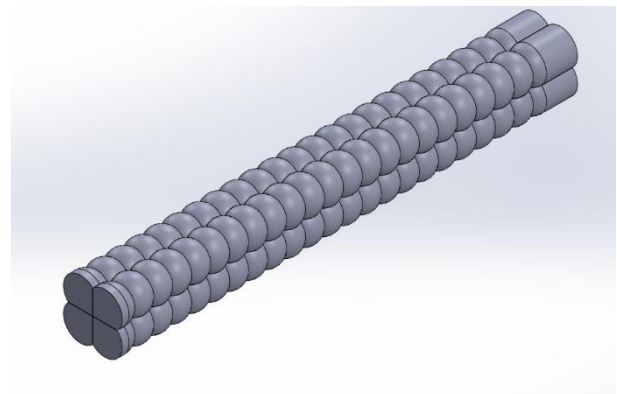


Figure 1. SolidWorks assembly of the modular Pneu-Net soft pneumatic actuator

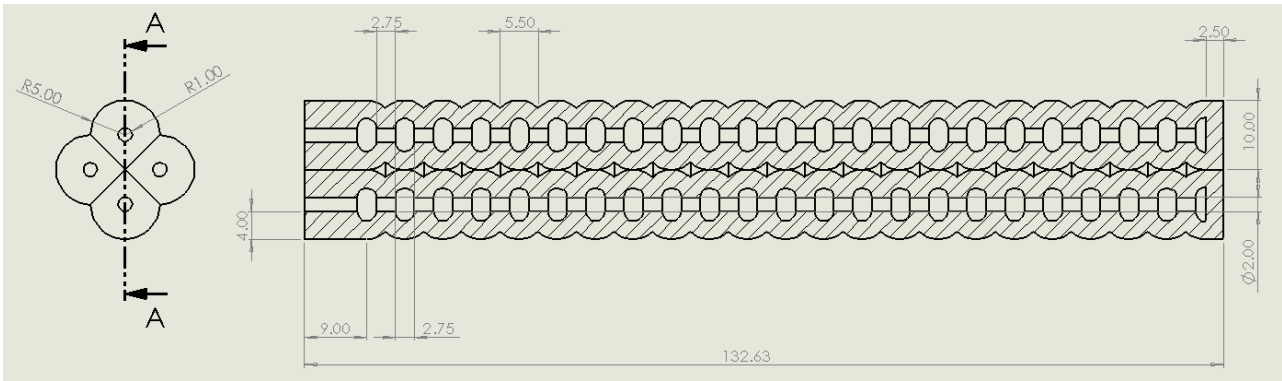


Figure 2. Engineering drawing of the 2.5 mm wall-thickness actuator configuration.

5. Finite Element Modeling

5.1 Analysis Setup

The STEP geometry was imported into Abaqus/CAE and prepared for nonlinear finite element analysis. Geometric continuity was checked after import, and the actuator was treated as a single analysis part for subsequent material assignment, boundary-condition definition, loading, and meshing.

The simulations were performed in Abaqus/Standard using a Static General analysis procedure. Geometric nonlinearity was enabled (NLGEOM = ON) to account for the large deformations expected during pneumatic actuation. Automatic incrementation was used during the analysis to support convergence as the actuator underwent progressively larger deformation.

5.2 Material Model

The actuator material was represented using a second-order Yeoh hyperelastic constitutive model to capture the nonlinear response of the elastomer under large deformation. The material parameters were derived from the literature-based values used for the reference actuator model [1], converted to a consistent SI unit system, and implemented in Abaqus following the Yeoh formulation [8].

In the final model, the Yeoh coefficients were defined as $C_{10} = 82,020$ Pa and $C_{20} = 96.9$ Pa. These values correspond to the literature-reported coefficients after conversion from MPa to Pa. No independent experimental material calibration or curve fitting was performed as part of this project.

Because the elastomer was treated as nearly incompressible, a hybrid finite element formulation was employed to avoid volumetric locking and improve numerical robustness. This choice is commonly required for incompressible or nearly incompressible hyperelastic materials in large-deformation finite element analysis.

5.3 Boundary Conditions, Loading, and Meshing

The actuator base, corresponding to the inlet-side attachment region, was constrained using an Encastre boundary condition, fixing all available displacement degrees of freedom at the base. Internal pressure was applied to the pneumatic chamber surfaces to generate the bending response of the actuator.

Pressure was introduced progressively during the nonlinear Static General analysis rather than being applied instantaneously. This loading strategy was used to support numerical convergence while capturing the strongly nonlinear response associated with hyperelastic material behavior and large deformation.

The complex internal chamber geometry was discretized using a free tetrahedral mesh. Quadratic hybrid tetrahedral elements (C3D10H) were employed because the hybrid formulation is appropriate for the nearly incompressible hyperelastic material used in the model. The resulting mesh was used for all subsequent displacement, deformation, and strain-related evaluations.

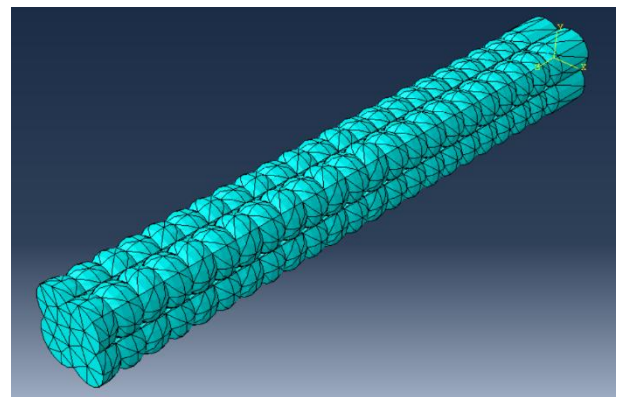


Figure 3. Tetrahedral finite element mesh of the actuator model in Abaqus/CAE.

6. Simulation Cases and Results

6.1 Simulation Matrix

The numerical study compared the pressure-driven response of the 2.0 mm and 2.5 mm wall-thickness configurations over multiple internal-pressure levels. The investigated cases were selected to capture the progression from relatively small deformation to increasingly pronounced bending while enabling direct comparison between the two actuator geometries.

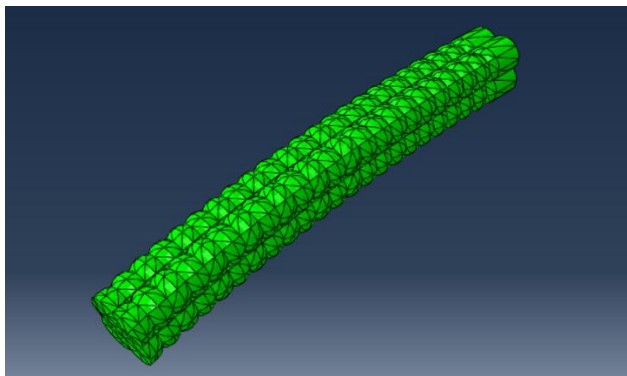
Wall Thickness	Investigated Pressure Cases
2.0 mm	10, 20, 35, 50 kPa
2.5 mm	10, 20, 35, 50, 70 kPa

Table 1. Investigated pressure cases for the two wall-thickness configurations.

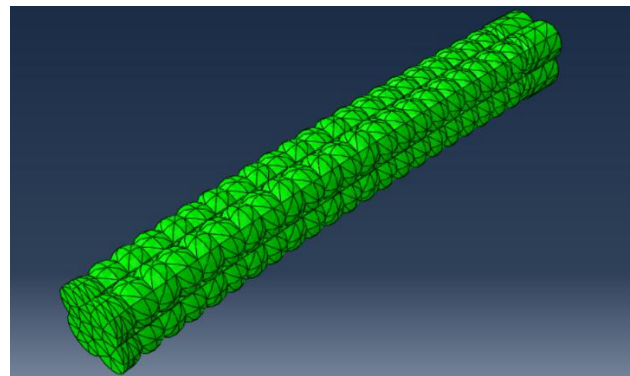
6.2 Pressure-Induced Deformation

Increasing internal pressure produced progressively larger actuator bending in both wall-thickness configurations. The thinner 2.0 mm model exhibited a more compliant response, while the 2.5 mm configuration showed comparatively reduced deformation under the same pressure level. This trend became increasingly evident as the applied pressure increased.

To visualize the effect of wall thickness, the deformed configurations were compared at common pressure levels. Representative results at 20 kPa and 35 kPa are shown below.

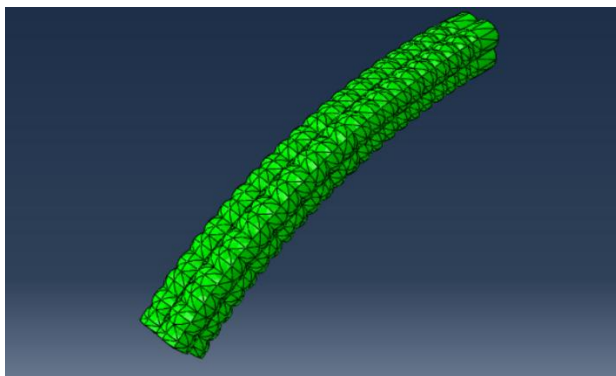


(a) 2.0 mm

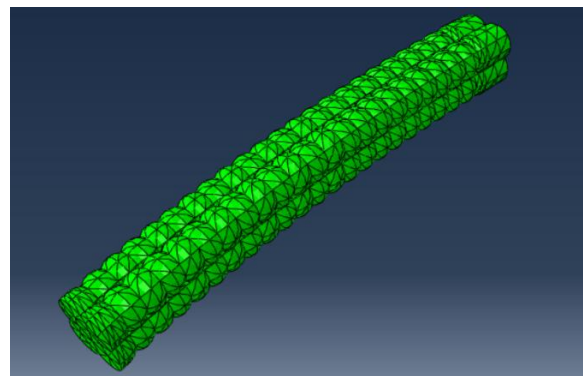


(b) 2.5 mm

Figure 4. Comparison of deformed actuator configurations at 20 kPa: (a) 2.0 mm wall thickness and (b) 2.5 mm wall thickness.



(a) 2.0 mm



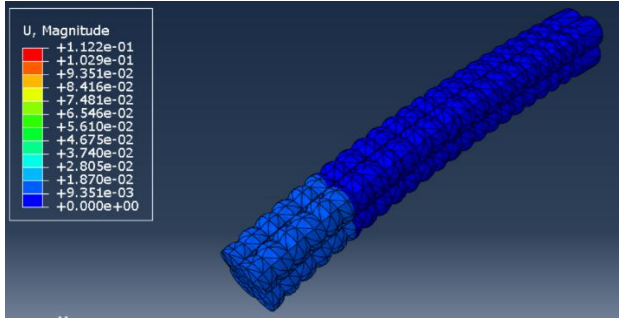
(b) 2.5 mm

Figure 5. Comparison of deformed actuator configurations at 35 kPa: (a) 2.0 mm wall thickness and (b) 2.5 mm wall thickness.

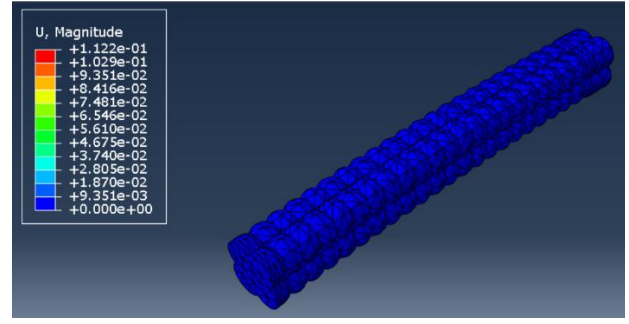
6.3 Displacement Magnitude

Displacement-magnitude contours were used to examine the spatial distribution of deformation along the actuator. At a common pressure level, the two wall-thickness configurations exhibited different deformation intensities, consistent with the greater compliance observed for the thinner actuator.

Figure 6 compares the displacement-magnitude fields of the 2.0 mm and 2.5 mm configurations at 20 kPa. The contours illustrate how the deformation develops along the actuator length and provide a qualitative comparison of the two geometries under the same applied pressure.



(a) 2.0 mm



(b) 2.5 mm

Figure 6. Displacement-magnitude contours at 20 kPa for (a) the 2.0 mm and (b) the 2.5 mm wall-thickness configurations.

6.4 Pressure–Response Trends

The simulation results exhibited a clear increase in actuator response with increasing internal pressure. The thinner-wall configuration generally showed a stronger deformation response, while the thicker configuration exhibited a comparatively stiffer behavior. This difference became more pronounced as the pressure level increased.

The reported pressure–strain curves provide an additional comparison of the two actuator configurations. Although the absolute strain response depends on the selected output measure and extraction location, the curves illustrate the overall sensitivity of the actuator behavior to both internal pressure and wall thickness.

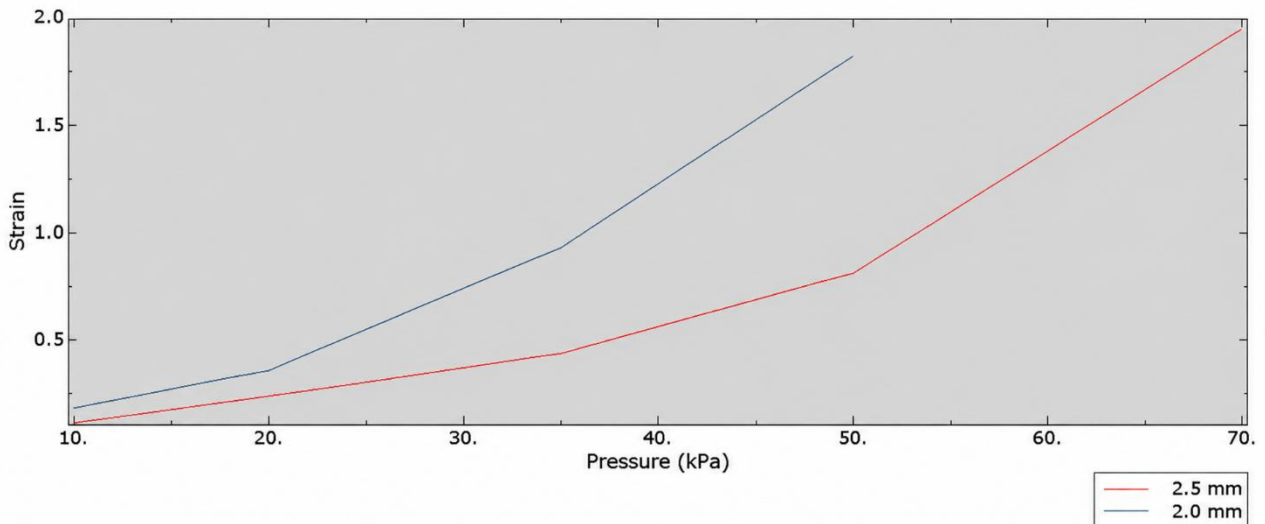


Figure 7. Reported pressure–strain response of the 2.0 mm and 2.5 mm actuator configurations.

6.5 Key Findings

The numerical results consistently showed that actuator deformation increased with internal pressure and that wall thickness had a clear influence on the resulting bending response. Across comparable pressure levels, the 2.0 mm configuration exhibited greater compliance and more pronounced deformation, whereas the 2.5 mm configuration produced a comparatively stiffer response.

The displacement-magnitude contours and pressure–strain trends further demonstrated the sensitivity of the actuator response to geometric thickness. These results highlight the importance of wall thickness as a design parameter in

balancing actuator flexibility and mechanical robustness.

Overall, the finite element study provided a consistent qualitative comparison between the two actuator configurations and established a computational basis for subsequent design refinement and future experimental validation.

7. Discussion

The results demonstrate the strong influence of wall thickness on the mechanical response of the modular Pneu-Net actuator. Reducing the wall thickness increased actuator compliance and produced larger pressure-induced deformation, while the thicker configuration exhibited a

comparatively stiffer response. This behavior reflects the expected trade-off between flexibility and structural robustness in soft pneumatic actuator design.

From a design perspective, a more compliant actuator may be advantageous when large bending deformation is required at relatively low pressure. In contrast, increasing wall thickness can provide a more restrained response and may be beneficial when greater geometric stability is desired. The appropriate thickness therefore depends on the intended balance between deformation capability, pressure range, and mechanical reliability.

The numerical model also highlights the importance of nonlinear material and geometric modeling in soft robotics. The use of a hyperelastic constitutive model, large-deformation kinematics, and hybrid finite elements was necessary to represent the strongly nonlinear behavior of the actuator. These modeling choices are particularly relevant for elastomeric pneumatic systems in which conventional linear assumptions would not adequately capture the observed response.

Although the actuator architecture is relevant to assistive and rehabilitation-oriented applications, the present work was limited to computational modeling and did not include fabrication, experimental validation, or human-subject testing. The results should therefore be interpreted as a preliminary design study and a basis for future experimental investigation rather than as validated device-level performance.

8. Conclusion

This project presented the CAD modeling and nonlinear finite element analysis of a modular Pneu-Net soft pneumatic actuator inspired by a previously reported bio-inspired design. The actuator geometry was developed in SolidWorks and analyzed in Abaqus/Standard using a second-order Yeoh hyperelastic material model, large-deformation kinematics, and quadratic hybrid tetrahedral elements.

Two wall-thickness configurations, 2.0 mm and 2.5 mm, were investigated under multiple internal-pressure cases. The simulations showed a clear pressure-dependent bending response and

demonstrated that wall thickness plays an important role in actuator compliance, deformation intensity, and numerical behavior. The thinner configuration exhibited greater flexibility, while the thicker configuration produced a comparatively stiffer response.

Overall, the study illustrates the value of nonlinear finite element analysis as a practical tool for preliminary design evaluation of soft pneumatic actuators. The developed workflow can support future design refinement, experimental validation, and further investigation of soft actuators intended for assistive and rehabilitation-oriented robotic applications.

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